

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VII – Independently Audited English Source-Aligned Checkpoint

Twelve bounded units; corrected French lines 2274–2498

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Checkpoint state. This is a bounded, coherent reflowed assembly of the mathematical bodies and source notes of twelve independently reviewed units. An independent cumulative mathematical, integration, build, render, machine-ledger, privacy, and package audit has passed for this exact checkpoint. It is not a critical edition or a claim of full-SGA 2 completion. Publication authorization and archival deposit remain decisions for the archive owner.

Evidence boundary. The corrected French arXiv TeX is the translation authority; the direct French PDF is the page and visual-reading control. The assembled body occupies original printed pp. 77–83, physical source-PDF pp. 69–74, and recomposed running pp. 61–66. Boundary blanks through line 2499 and the layout command at line 2500 have been checked; Exposé VIII begins at line 2501 and is excluded. This checkpoint is not a critical edition or certification of the French source.

Unit	Scope	French lines	Printed pp.	Physical pp.	Running pp.	Bounded-unit review
C01	Opening; Lemma 1.1 statement	2274–2292	77	69	61	reviewed unit
C02	Lemma 1.1 proof	2294–2320	77–78	69–70	61–62	reviewed unit
C03	Proposition 1.2 statement	2322–2349	78–79	70	62	reviewed unit
C04	Proposition 1.2 induction; Lemma 1.3 statement	2351–2375	79	70–71	62–63	reviewed unit
C05	Lemma 1.3 proof; Proposition 1.2 close	2377–2384	80	71	63	reviewed unit
C06	Corollary 1.4 and proof	2386–2410	80	71	63	reviewed unit
C07	Transition; Lemma 1.5 and proof	2412–2425	81	71–72	63–64	reviewed unit
C08	Lemma 1.6; Proposition 1.7	2427–2437	81	72	64	reviewed unit
C09	Section 2 opening; Proposition 2.1 statement	2439–2457	82	72	64	reviewed unit
C10	Proposition 2.1 proof	2459–2482	82–83	73	65	reviewed unit
C11	Definition $D(P)$; Lemma 2.2 and proof	2484–2493	83	73	65	reviewed unit
C12	Proposition 2.3 and proof; Exposé VII close	2495–2498	83	73–74	65–66	reviewed unit

Exposé VII. Vanishing Criteria and Coherence Conditions for the Sheaves $\mathcal{E}xt_Y^i(F, G)$

1. Study for $i < n$

We first prove a lemma.

Lemma 1.1. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X , and let G be a quasi-coherent $\mathcal{O}_X$ -module. Suppose that, for every coherent $\mathcal{O}_X$ -module F whose support is contained in Y , one has*

$$\mathcal{E}xt^{n-1}(F, G) = 0.$$

Then, for every coherent $\mathcal{O}_X$ -module F and every closed subset Z of X such that $Y \supset \text{Supp } F \cap Z$, one has an isomorphism

$$\mathcal{E}xt_Z^n(F, G) \cong \mathcal{H}om(F, \mathcal{H}_Y^n(G)).$$

Proof of Lemma 1.1

We first observe that

$$\mathcal{E}xt_Z^i(F, G) = \mathcal{E}xt_{Z \cap \text{Supp } F}^i(F, G).$$

(This is immediate; cf. Exposé VI.) We first prove the assertion for $Z = X$, so that $\text{Supp } F \subset Y$. On the category of coherent $\mathcal{O}_X$ -modules whose support is contained in Y , the functor

$$F \longmapsto \mathcal{E}xt^n(F, G)$$

is left exact. By (IV, 1.3), it is represented by

$$I = \varinjlim_k \mathcal{E}xt^n(\mathcal{O}_X/\mathcal{I}^{k+1}, G),$$

where $\mathcal{I}$ is a defining ideal of Y . By (II, 6), one has

$$\mathcal{H}_Y^n(G) \cong \varinjlim_k \mathcal{E}xt^n(\mathcal{O}_X/\mathcal{I}^{k+1}, G).$$

This proves the assertion when $Z = X$. Again by (VI, 2.3),

$$\mathcal{E}xt_Z^n(F, G) \cong \varinjlim_k \mathcal{E}xt^n(F/\mathcal{J}^{k+1}F, G),$$

where $\mathcal{J}$ is a defining ideal of Z . The support of $F/\mathcal{J}^{k+1}F$ is contained in Y whenever $Z \cap \text{Supp } F \subset Y$. What we have just proved therefore gives

$$\mathcal{E}xt_Z^n(F, G) \cong \varinjlim_k \mathcal{H}om(F/\mathcal{J}^{k+1}F, \mathcal{H}_Y^n(G)).$$

It remains to show that the natural homomorphism

$$\varinjlim_k \mathcal{H}om(F/\mathcal{J}^{k+1}F, \mathcal{H}_Y^n(G)) \longrightarrow \mathcal{H}om(F, \mathcal{H}_Y^n(G))$$

is an isomorphism when $Z \cap \text{Supp } F \subset Y$. Since X admits a cover by Noetherian affine open subsets, we reduce to the case where X is affine Noetherian. Then $F(X)$ is a finitely generated $\mathcal{O}_X(X)$ -module and $\text{Supp } \mathcal{H}_Y^n(G) \subset Y$. Hence every homomorphism

$$u : F(X) \longrightarrow \mathcal{H}_Y^n(G)(X)$$

is annihilated by a power of $\mathcal{I}$, and therefore by a power of $\mathcal{J}$. This proves the lemma.

Proposition 1.2. *Let X be a locally Noetherian prescheme, let Y be a closed subset of X , let G be a quasi-coherent $\mathcal{O}_X$ -module, and let n be an integer. For any closed subsets Z and S of X such that $Z \cap S = Y$, the following conditions are equivalent:*

(i) $\mathcal{H}_Y^i(G) = 0$ for $i < n$;

(ii) there exists a coherent $\mathcal{O}_X$ -module F with support S such that

$$\mathcal{E}xt_Z^i(F, G) = 0 \quad \text{for } i < n;$$

(iii) for every coherent $\mathcal{O}_X$ -module F whose support is contained in S (that is, $\text{Supp } F \cap Z = \text{Supp } F \cap Y$), one has

$$\mathcal{E}xt_Z^i(F, G) = 0 \quad \text{for } i < n;$$

(iv) for every coherent $\mathcal{O}_X$ -module F , one has

$$\mathcal{E}xt_Y^i(F, G) = 0 \quad \text{for } i < n.$$

Moreover, if these conditions hold, then for every coherent $\mathcal{O}_X$ -module F and every closed subset Z' of X such that $Z' \cap \text{Supp } F = Y \cap \text{Supp } F$, one has isomorphisms

$$\mathcal{E}xt_Z^n(F, G) \cong \mathcal{E}xt_Y^n(F, G) \cong \mathcal{H}om(F, \mathcal{H}_Y^n(G)).$$

Source note. The quantified subset in the preceding sentence is Z' , but both the corrected French TeX and the direct compiled page print Z in the first term of the final display. The proof immediately below uses Z' . Both source controls confirm these facts; Z' is probably intended, but the edition decision remains unresolved. This reader therefore preserves printed Z and does not silently emend the authority.

Proof of Proposition 1.2: induction

We argue by induction. The proposition is immediate for $n < 0$. Suppose that it has been proved for $n < q$. If one of the conditions holds for $n = q$ and for two subsets Z and S as in the statement, then the induction hypothesis gives, for every closed subset Z' of X and every coherent $\mathcal{O}_X$ -module F such that $Z' \cap \text{Supp } F = Y \cap \text{Supp } F$, isomorphisms

$$\mathcal{E}xt_{Z'}^{q-1}(F, G) \cong \mathcal{H}om(F, \mathcal{H}_Y^{q-1}(G)) \cong \mathcal{E}xt_Y^{q-1}(F, G). \quad (1.1)$$

Consequently:

(i) $\Rightarrow$ (iv) take $Z' = Y$ in (1.1);

(iv) $\Rightarrow$ (iii) take $Z' = Z$ in (1.1);

(iii) $\Rightarrow$ (ii) take $F = \mathcal{O}_S$;

(ii) $\Rightarrow$ (i) take $Z' = Z$ in (1.1). This gives $\mathcal{H}om(F, \mathcal{H}_Y^{q-1}(G)) = 0$. Moreover,

$$\text{Supp } \mathcal{H}_Y^{q-1}(G) \subset Y = Z \cap S \subset S = \text{Supp } F,$$

so the following lemma applies.

Lemma 1.3. *Let X be a prescheme, let P be a coherent $\mathcal{O}_X$ -module, and let H be a quasi-coherent $\mathcal{O}_X$ -module such that*

$$\mathcal{H}om(P, H) = 0 \quad \text{and} \quad \text{Supp } P \supset \text{Supp } H.$$

Then $H = 0$.

Proof of Lemma 1.3

It suffices to prove the lemma when X is affine, since the affine open subsets form a basis for the topology of X and the hypotheses are preserved under restriction to an open subset. In that case, we are reduced to a problem about A -modules, where $X = \text{Spec}(A)$. We apply the formula (valid under the sole hypothesis that M is of finite type):

$$\text{Ass Hom}_A(P, H) = \text{Supp } P \cap \text{Ass } H.$$

We know that

$$\text{Ass } H \subset \text{Supp } H \subset \text{Supp } P \quad \text{and} \quad \text{Ass Hom}_A(P, H) = \emptyset;$$

hence $\text{Ass } H = \emptyset$, and therefore $H = 0$.

Source note. The corrected French TeX and direct PDF print M in the finite-type parenthetical. Since the displayed formula involves P and no M is introduced, P is the high-confidence intended reading; no silent emendation is made while the French-authority disposition is pending.

Close of the proof of Proposition 1.2

To complete the proof of the proposition, it remains to observe that condition (iv) permits the application of Lemma 1.1.

Corollary 1.4. *Let G be a coherent Cohen–Macaulay $\mathcal{O}_X$ -module, and let $n \in \mathbb{Z}$. The conditions of Proposition 1.2 are equivalent to:*

$$(v) \quad \text{codim}(Y \cap \text{Supp } G, \text{Supp } G) \geq n.$$

Proof

Recall first that an $\mathcal{O}_X$ -module is said to be Cohen–Macaulay if, for every $x \in X$, the stalk G_x is a Cohen–Macaulay $\mathcal{O}_{X,x}$ -module; that is, for every $x \in S = \text{Supp } G$,

$$\text{depth } G_x = \dim G_x = \dim \mathcal{O}_{S,x}. \quad (1.2)$$

By Proposition III.3.3, condition (i) of Proposition 1.2 is equivalent to

$$\text{depth}_Y G = \inf_{x \in Y} \text{depth } G_x \geq n, \quad (1.3)$$

and hence also to

$$\text{depth}_Y G = \inf_{x \in Y \cap S} \text{depth } G_x \geq n,$$

since the depth of the zero module is infinite.

By definition,

$$\text{codim}(Y \cap S, S) = \inf_{x \in S \cap Y} \dim \mathcal{O}_{S,x};$$

the conclusion follows by applying formula (1.2).

We shall now prove a result that will allow us to deduce the coherence conditions we have in view from certain vanishing criteria.

Lemma 1.5. *Let X be a locally Noetherian prescheme. Let T^* be an exact contravariant ∂ -functor defined on the category of coherent $\mathcal{O}_X$ -modules and taking values in the category of $\mathcal{O}_X$ -modules. Let Y be a closed subset of X , and let $i \in \mathbb{Z}$. Suppose that, for every coherent $\mathcal{O}_X$ -module F whose support is contained in Y , the modules $T^i F$ and $T^{i-1} F$ are coherent. Let F be a coherent $\mathcal{O}_X$ -module. Then $T^i F$ is coherent if and only if $T^i F''$ is coherent, where*

$$F'' = F/\Gamma_Y(F).$$

Indeed, $F' = \Gamma_Y(F)$ is coherent because X is locally Noetherian. The long exact cohomology sequence of T^* then gives

$$T^{i-1} F' \longrightarrow T^i F'' \longrightarrow T^i F \longrightarrow T^i F',$$

where the two outer terms are coherent; the conclusion follows.

Lemma 1.6. *If F and G are coherent and $\text{Supp } F \subset Y$, then $\mathcal{E}xt_Y^i(F, G)$ is coherent.*

Indeed, $\mathcal{E}xt_Y^i(F, G)$ is isomorphic to $\mathcal{E}xt^i(F, G)$. More generally, this holds on any ringed space X : if Z is a closed subset containing $Y \cap \text{Supp } F$, then $\mathcal{E}xt_Z^i(F, G)$ is isomorphic to $\mathcal{E}xt_Y^i(F, G)$ (cf. Exposé VI).

Proposition 1.7. *Suppose that F and G are coherent, and put*

$$\text{Supp } F = S, \quad S' = \overline{S \cap (X - Y)}.$$

Suppose that $\text{depth } G_x \geq n$ for every $x \in Y \cap S'$. Then $\mathcal{E}xt_Y^i(F, G)$ is coherent for $i < n$.

Indeed, Lemma 1.6 allows us to apply Lemma 1.5 to

$$T^*(F) = \mathcal{E}xt_Y^*(F, G).$$

Putting $F'' = F/\Gamma_Y(F)$, we see that $\text{Supp } F'' = S'$. By Proposition III.3.3, the hypothesis on the depth of G ensures that $\mathcal{H}_{Y \cap S'}^i(G) = 0$ for $i < n$; Proposition 1.2 then gives $T^i F'' = 0$ for $i < n$, and the conclusion follows from Lemma 1.5.

2. Study for $i > n$

Let X be a locally Noetherian *regular* prescheme; that is, suppose that all its local rings are regular. Let Y be a closed subset of X , and let F and G be two coherent $\mathcal{O}_X$ -modules. Put

$$S = \text{Supp } F, \quad S' = \overline{S \cap (X - Y)},$$

and put

$$\begin{aligned} m &= \sup_{x \in Y \cap S} \dim \mathcal{O}_{X,x}, \\ n &= \sup_{x \in Y \cap S'} \dim \mathcal{O}_{X,x}. \end{aligned}$$

Then $n \leq m$.

Proposition 2.1. *In the situation described above:*

1. $\mathcal{E}xt_Y^i(F, G) = 0$ if $i > m$;
2. $\mathcal{E}xt_Y^i(F, G)$ is coherent if $i > n$.

Proof of Proposition 2.1

First note that $\mathcal{E}xt_Y^i(F, G)$ is coherent for every i when $\text{Supp } F \subset Y$. Moreover, putting $F'' = F/\Gamma_Y(F)$ as above, we see that $\text{Supp } F'' = S'$. Thus assertion (2) follows from assertion (1) and Lemma 1.5.¹

To prove assertion (1), first observe that

$$\mathcal{E}xt_Y^i(F, G) \cong \varinjlim_k \mathcal{E}xt^i(F/\mathcal{J}^k F, G),$$

where $\mathcal{J}$ is the ideal defining Y . On the other hand, Theorem 4.2.2 of A. Grothendieck, “Sur quelques points d’algèbre homologique,” *Tôhoku Mathematical Journal* **9** (1957), 119–221, shows that the $\mathcal{E}xt$ sheaves commute with the formation of stalks, at least when X is a locally Noetherian prescheme and the first argument is coherent. Since direct limits do likewise, for every $x \in X$ there are isomorphisms²

$$(\mathcal{E}xt_Y^i(F, G))_x \cong \varinjlim_k \text{Ext}_{\mathcal{O}_{X,x}}^i((F/\mathcal{J}^k F)_x, G_x).$$

Since

$$\text{Supp } \mathcal{E}xt_Y^i(F, G) \subset S \cap Y,$$

¹The French TeX and printed page cite Lemma 1.3. The coherence reduction used here is Lemma 1.5; Lemma 1.3 is the earlier $\mathcal{H}om$ /support lemma.

²The French TeX and printed stalk formula omit the superscript i on the local Ext term. The preceding direct-limit formula and the following vanishing display both require Ext^i .

it is enough to note that $x \in Y \cap S$ implies $\dim \mathcal{O}_{X,x} \leq m$, and hence

$$\mathrm{Ext}_{\mathcal{O}_{X,x}}^i((F/\mathcal{I}^k F)_x, G_x) = 0 \quad \text{if } i > m,$$

because the global cohomological dimension of a regular local ring is equal to its dimension.³

Let X be a locally Noetherian prescheme. For every subset P of X , set

$$D(P) = \{\dim \mathcal{O}_{X,p} \mid p \in P\}.$$

Lemma 2.2

If P is the underlying space of a connected subprescheme of X , then $D(P)$ is an interval.⁴

Indeed, let a and b belong to $D(P)$, corresponding to points p and q of P . We show that there is a sequence of points of P ,

$$p = p_1, \dots, p_n = q,$$

such that, for $1 \leq i < n$,

$$|\dim \mathcal{O}_{X,p_i} - \dim \mathcal{O}_{X,p_{i+1}}| = 1.$$

It will follow that $D(P)$ contains the interval $[a, b]$.⁵

For this, observe that p and q can be joined by a chain of irreducible components of P in which two successive components meet. We are reduced to the case where p is the generic point of an irreducible component Q of P , and where $q \in Q$, hence $q \supset p$ as ideals of $\mathcal{O}_q$; there the assertion follows immediately from the definition of dimension.

³Cf. EGA 0_{IV}, 17.3.1.

⁴The French TeX and printed witness read “of A .” The ambient prescheme introduced immediately before is X , and no object A occurs in this argument. The reader therefore uses X transparently.

⁵The French TeX and printed witness introduce the numerical endpoints as “ a and B ” and then print the interval $[p, q]$, although p and q are points of P . The reader normalizes the two numerical endpoints to a, b and the interval to $[a, b]$. The absolute-value bars are present in the current French TeX and printed witness and are preserved.

Proposition 2.3

Let X be a locally Noetherian regular prescheme, let Y be a closed subset of X , and let F be a coherent $\mathcal{O}_X$ -module. Put

$$P = Y \cap \overline{\text{Supp } F \cap (X - Y)}.$$

Let $n \in \mathbb{Z}$, and suppose that $n \notin D(P)$. Then $\mathcal{E}xt_Y^n(F, \mathcal{O}_X)$ is coherent.⁶

Proof

The conclusion is local, and the hypotheses are preserved under restriction to an open subset. Since P is closed, it is locally Noetherian and hence locally connected. We may therefore suppose that X is affine and Noetherian and that P is connected.

If $P = \emptyset$, the depth hypothesis of Proposition 1.7 with threshold $n + 1$ is vacuous, and the conclusion follows. Assume henceforth that $P \neq \emptyset$. By Lemma 2.2, $D(P)$ is an interval. Since $n \notin D(P)$, exactly one of the following alternatives holds:⁷

$$n > d \quad \text{for every } d \in D(P), \quad \text{or} \quad n < d \quad \text{for every } d \in D(P).$$

In the first case, put $m = \sup D(P)$. Then $n > m$, and Proposition 2.1, applied with $G = \mathcal{O}_X$, shows that $\mathcal{E}xt_Y^n(F, \mathcal{O}_X)$ is coherent.

In the second case, for every $x \in P$ regularity gives

$$n < \dim \mathcal{O}_{X,x} = \text{depth } \mathcal{O}_{X,x},$$

and hence $\text{depth } \mathcal{O}_{X,x} \geq n + 1$. Proposition 1.7 with threshold $n + 1$ therefore gives coherence in every degree $i < n + 1$, in particular in degree $i = n$.

⁶The French TeX and printed statement omit the closure and read $P = Y \cap \text{Supp } F \cap (X - Y)$, which is empty literally. Section 2 defines $S' = \overline{\text{Supp } F \cap (X - Y)}$, and Propositions 2.1 and 1.7 both use the locus $Y \cap S'$. The reader restores that uniquely supported closure.

⁷The French TeX and print write $D(P) = [a, b[$ and then treat only $n > b$ and $n < a$, leaving the allowed case $n = b$ uncovered. The print does not decide whether the intended one-character repair was a closed upper endpoint or a non-strict upper inequality. The reader states the order alternatives forced by $n \notin D(P)$ and Lemma 2.2 and therefore does not guess that glyph.